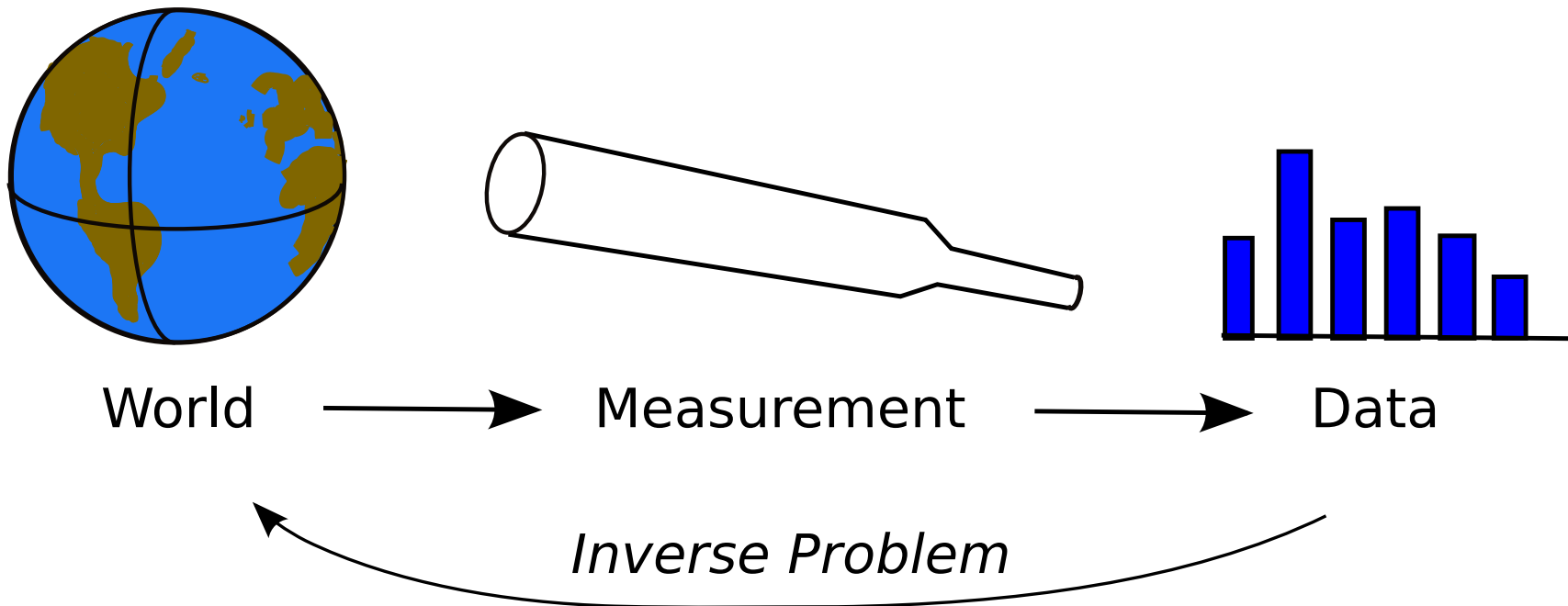


# Mathematics of Machine Learning

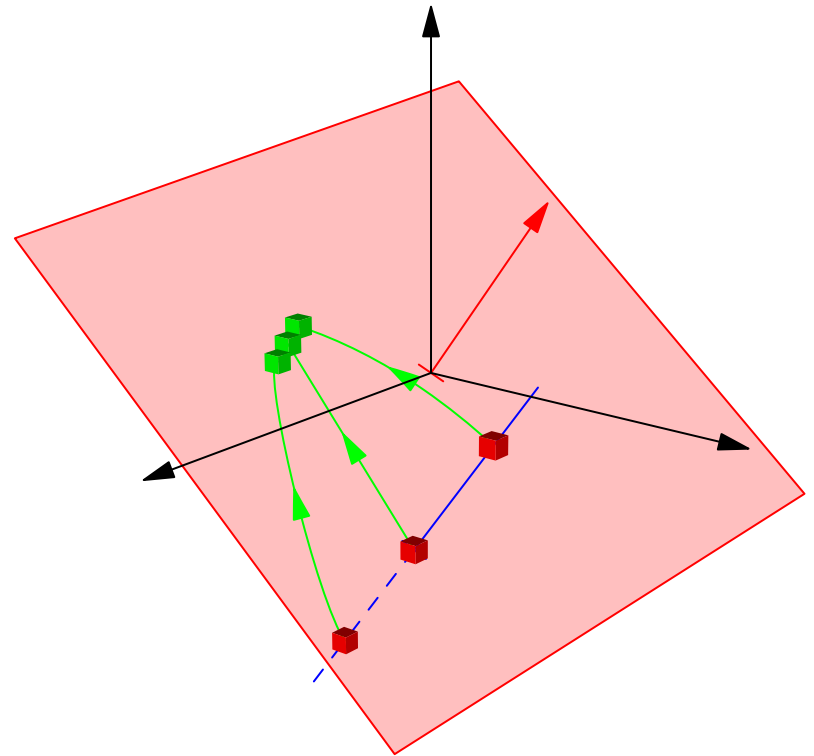
## *Understand Mappings*



*Least Squares, Under-constrained Systems, Ill-conditioning*

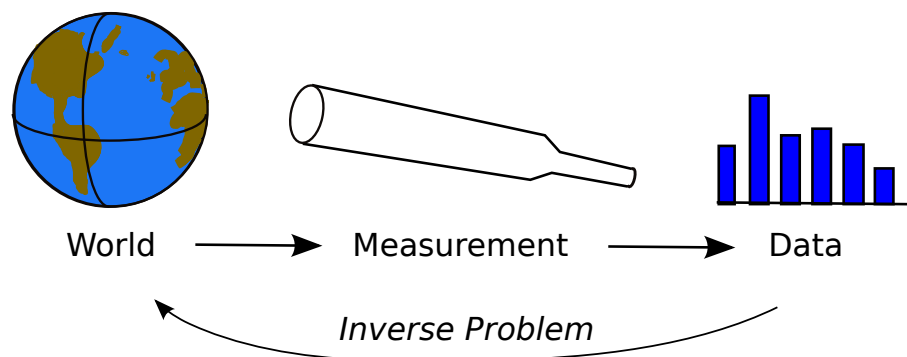
# Outline

1. **Least Squares**
2. Under-constrained Systems
3. Ill-conditioning



# Transforming Data

- In the last lecture we spent time developing a sophisticated view of vector spaces and operators
- At a mathematical level machine learning can be viewed as performing an inverse mapping



- Although our mappings are not necessarily linear in either direction we learn a lot by understanding linear operators

# Inverse Problems

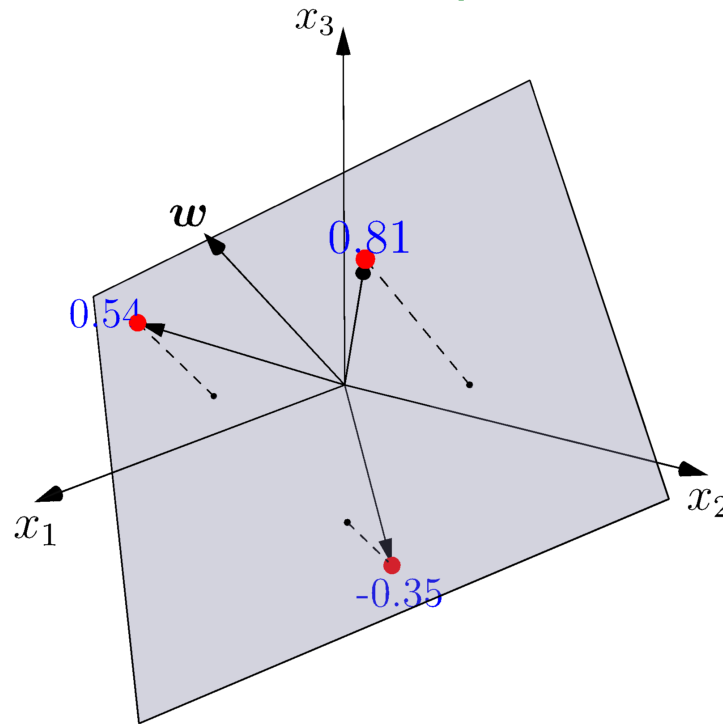
- Given  $m$  observations  $\{(\mathbf{x}_k, y_k) | k = 1, \dots, m\}$ , find the  $p$  unknowns  $\mathbf{w} = (w_1, w_2, \dots, w_p)^\top$  such that  $\mathbf{x}_k^\top \mathbf{w} = y_k$  ■
- Define the *design matrix* as the matrix of feature vectors

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mp} \end{pmatrix} \quad \blacksquare$$

- and the target vector  $\mathbf{y} = (y_1, y_2, \dots, y_m)^\top$  ■
- Then  $\mathbf{y} = \mathbf{X}\mathbf{w}$  and if  $m = p$  (and  $\mathbf{X}$  is invertible)  $\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$  ■

# Linear Regression

- $x_k^T w$  depends on distance from the plane orthogonal to  $w$  ■



- If  $m > p$  then  $X$  isn't square so doesn't have an inverse ■
- Worse, unless the data is exact,  $y \approx Xw \Rightarrow$  no exact "solution" ■
- Problem solved by Gauss to predict the orbit of the asteroid Ceres ■

# Linear Least Squares

- The error of input pattern  $\mathbf{x}_k$  is

$$\epsilon_k = \mathbf{x}_k^\top \mathbf{w} - y_k$$

- We can define the error vector

$$\boldsymbol{\epsilon} = \mathbf{X}\mathbf{w} - \mathbf{y}$$

(note that  $\epsilon_k = \mathbf{x}_k^\top \mathbf{w} - y_k$ )

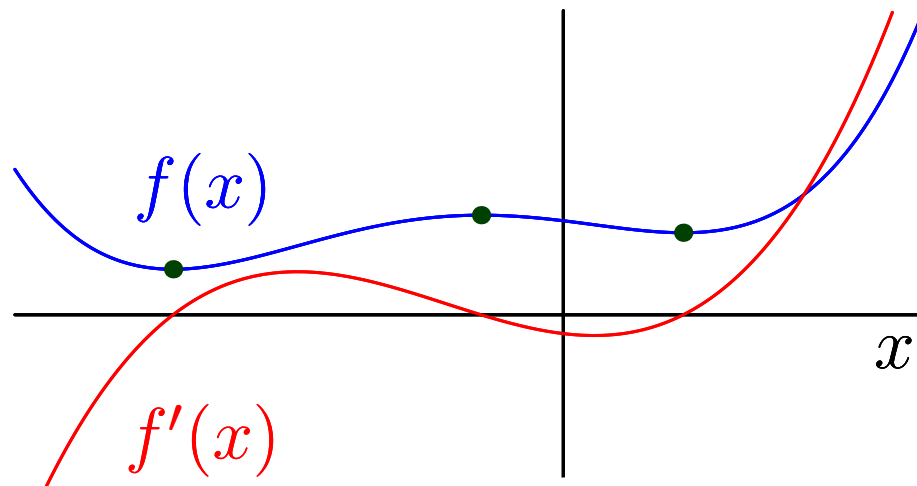
- The squared loss

$$L(\mathbf{w}|\mathcal{D}) = \sum_{k=1}^m (\mathbf{x}_k^\top \mathbf{w} - y_k)^2 = \sum_{k=1}^m \epsilon_k^2 = \|\boldsymbol{\epsilon}\|^2$$

- Minimising this loss is known as the least squares problem

# Finding a Minimum

- At a minimum of a one-dimensional function,  $f(x)$ , we have  $f'(x) = 0$



- At a minimum of an  $n$ -dimensional function  $f(\mathbf{x})$  we have the set of equations

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0 \quad \forall i = 1, \dots, n$$

# Least Squares Solution

- The least squares solution is given by

$$\begin{aligned}\nabla L(\mathbf{w}|\mathcal{D}) &= \nabla \|\epsilon\|^2 = \nabla \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 \\ &= \nabla (\mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}) \\ &= 2(\mathbf{X}^\top \mathbf{X} \mathbf{w} - \mathbf{X}^\top \mathbf{y}) = 0\end{aligned}$$

- Or

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^+ \mathbf{y}$$

- $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is known as the pseudo-inverse



# Missing Bits of the Mathematics

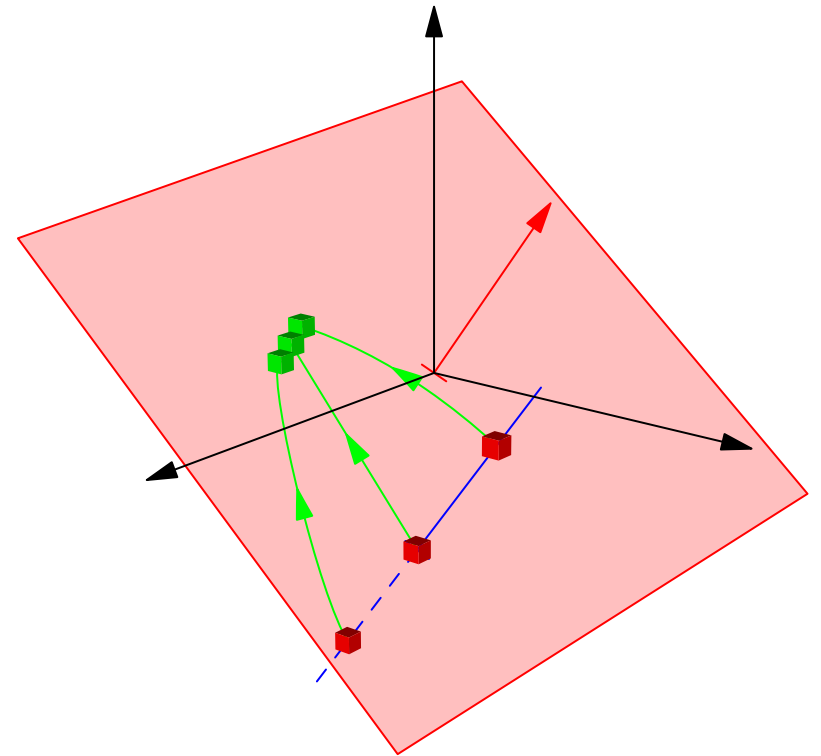
- Note that  $\|a\|^2 = a^\top a = \sum_i a_i^2$

$$\begin{aligned}\|Xw - y\|^2 &= (Xw - y)^\top (Xw - y) = (w^\top X^\top - y^\top)(Xw - y) \\ &= w^\top X^\top Xw - 2w^\top X^\top y + y^\top y\end{aligned}$$

- Where we have used  $w^\top X^\top y = y^\top Xw$ ,  $\sum_{i,j} w_i X_{ji} y_j = \sum_{i,j} y_i X_{ij} w_j$
- Also  $\nabla w^\top M w = M w + M^\top w = (M + M^\top) w$
- If  $M = M^\top$  (i.e.  $M$  is symmetric) then  $\nabla w^\top M w = 2M w$
- As  $(AB)^\top = B^\top A^\top$  then  $(X^\top X)^\top = X^\top X^{\top\top} = X^\top X$  and  $X^\top X$  is symmetric

# Outline

1. Least Squares
2. **Under-constrained Systems**
3. Ill-conditioning

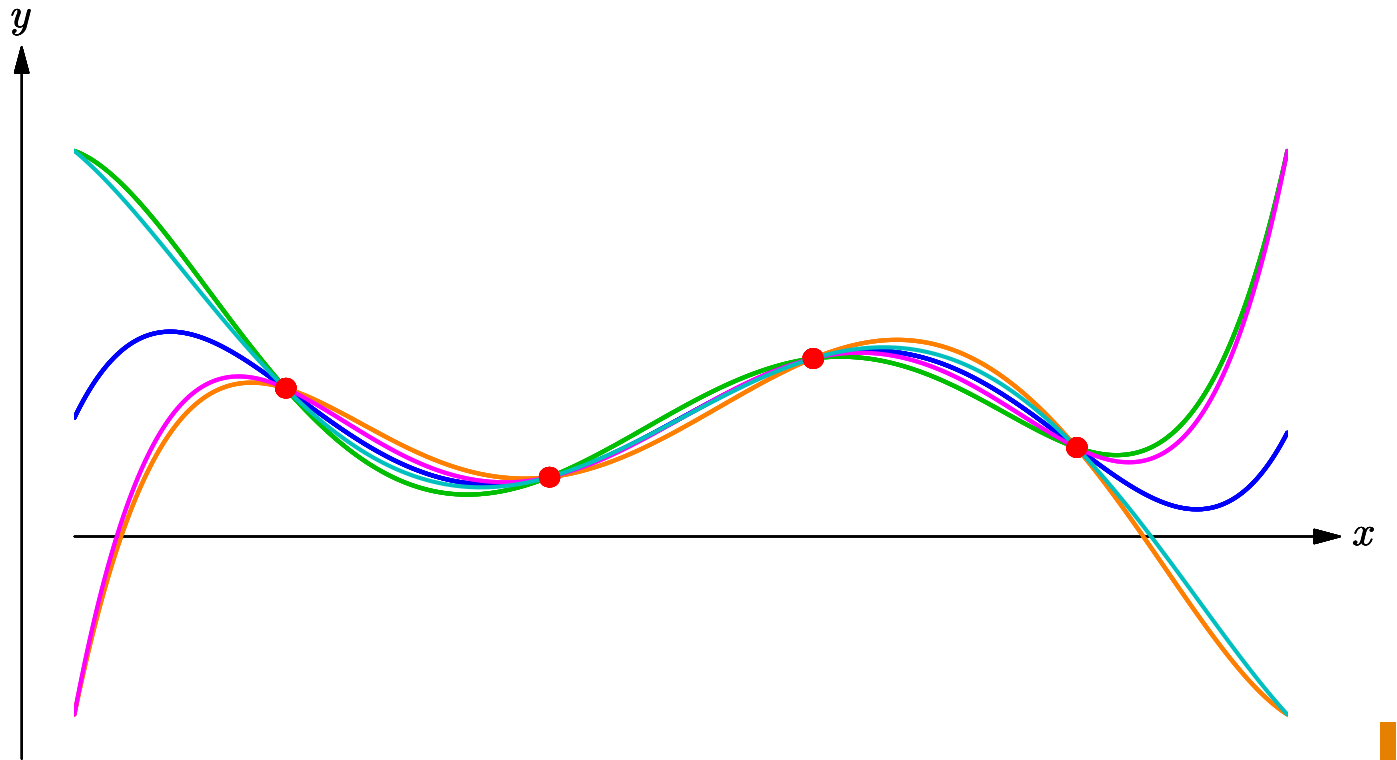


# Solving Inverse Problems

- Gauss showed us how to solve **over-constrained** problems (we have more observations than parameters)■
- We seek a solution which isn't necessarily exact but minimises an error■
- But, what if we have more parameters than observations?■
- That is, we are **under-constrained**■
- Note that in some directions you might be over-constrained and in other directions under-constrained■
- This is very typical of most machine learning problems■

# Under-constrained Systems

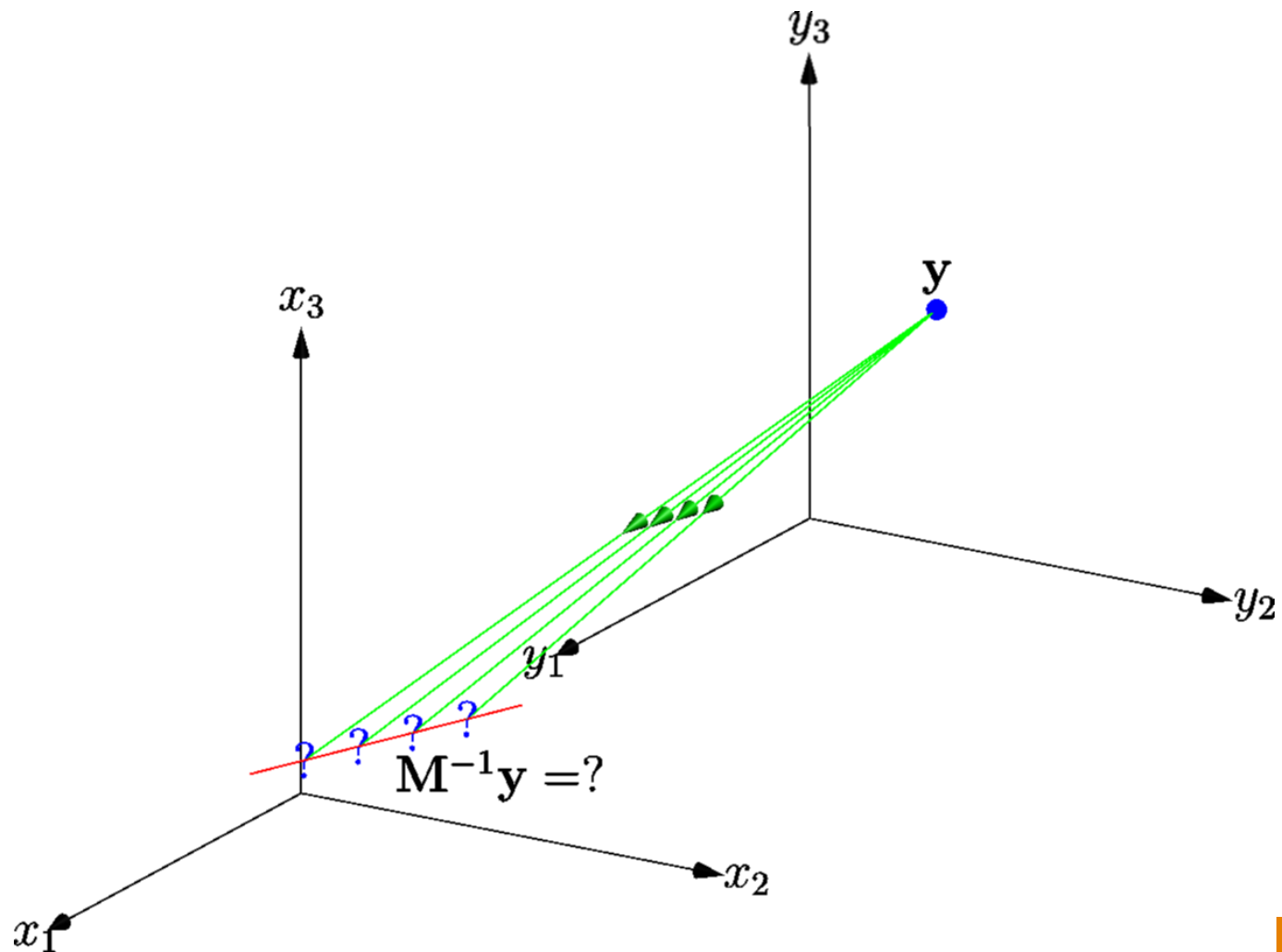
- If we have fewer data points than parameters then there will be multiple solutions, e.g. fitting  $y = w_0 + w_1x + \dots + w_5x^5$  to 4 points



- Every curve fits the data exactly, so which one should we choose?

# What is the Inverse?

- Many points can map to the same point



# Under-constrained Systems

- The system is **under-constrained**■
- We have more unknowns than equations■
- The inverse is not unique■
- Solving the inverse problem ( $w = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ ) is said to be **ill-posed**■
- The inverse  $(\mathbf{X}^T \mathbf{X})^{-1}$  doesn't exist■
- If we have a complicated learning machine and not sufficient data we often end up with an ill-posed inverse problem (there are lots of sets of parameters that explain the data)■

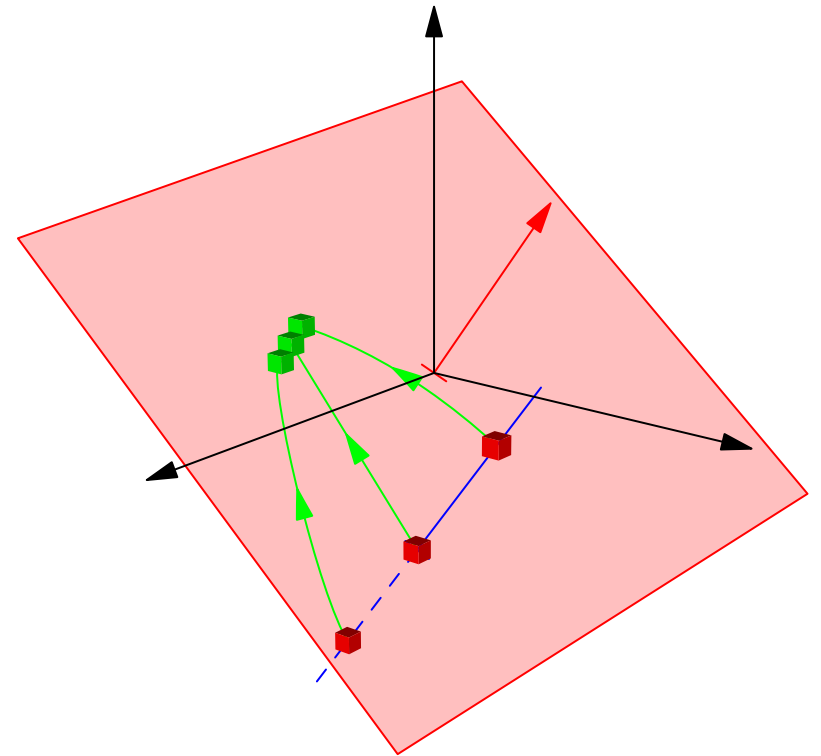
# Handling Under-constrained Systems

- The issue with under-constrained systems is not that there is no solution
- There is a whole family of solutions, so which one should we pick
- William of Ockham in the 14th century postulated that the “simplest solution” is likely to be the correct solution
- How we determine the simplest solution is a bit of a black art
- However, for linear regression, adding an  $L_2$  regulariser makes the solution unique

$$L_{full} = L_{fit}(\mathbf{w}) + \eta \|\mathbf{w}\|_2^2$$

# Outline

1. Least Squares
2. Under-constrained Systems
3. **Ill-conditioning**

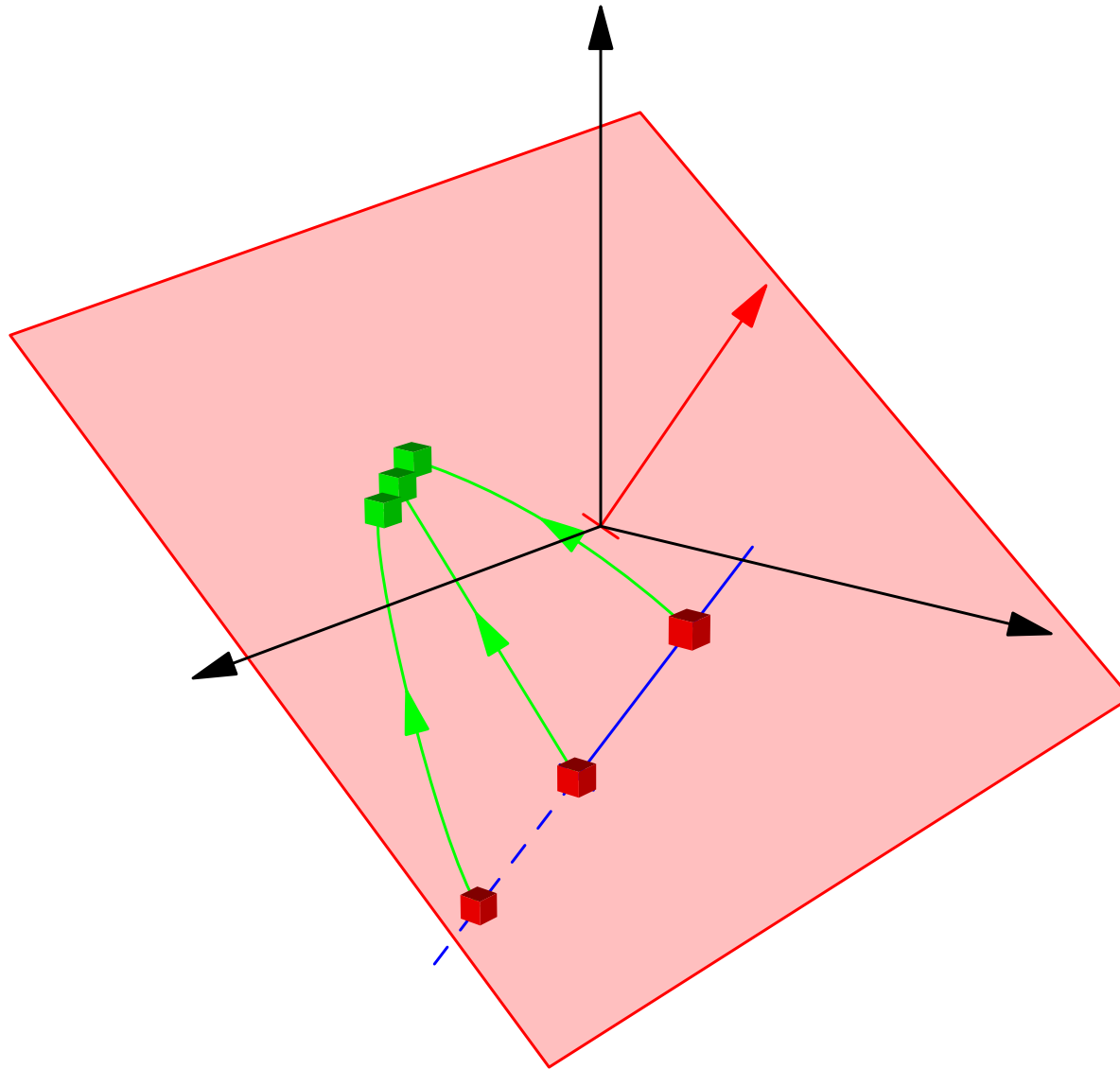




# III-Conditioning

- Singular matrices are rare (although they occur when we don't have enough data), but matrices that are close to being singular are common■
- If a matrix is close to singular it is ill-conditioned■
- Ill-conditioned matrices have some eigenvalues that are tiny compared with the largest■
- All points get squashed towards a lower-dimensional subspace (a plane in 3-D)■
- Large matrices are very often ill-conditioned■

# III-Conditioned Matrices



# A Worked Example

- Two data points and two features that are almost the same

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1.001 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 2 \\ 2.001 \end{pmatrix} \Rightarrow \mathbf{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- Change  $y_2$  by 0.001 (i.e. 0.05%)

$$\mathbf{y} = \begin{pmatrix} 2 \\ 2.002 \end{pmatrix} \Rightarrow \mathbf{w} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

- The prediction for a new input  $\mathbf{x} = (1,0)^\top$  jumps from 1 to 0
- The eigenvalues of  $\mathbf{X}$  are about 2 and 0.0005
- $\mathbf{w}$  moved along  $(-1,1)^\top$ , the direction with the tiny eigenvalue, which the data hardly constrains

# III-Conditioning in ML

- Ill-conditioning in machine learning occurs when a very small change in the learning data causes a large change in the predictions of the learning machine■
- In linear regression the matrix  $\mathbf{X}^T\mathbf{X}$  is typically ill-conditioned when we have roughly as many data points as parameters■
- Much of machine learning is concerned with making learning machines better conditioned■
- Adding regularisers is one approach to achieve this■

# Where Does Ill-Conditioning Come From?

- Too little data: about as many data points as parameters■
- Nearly collinear features, e.g. floor area and number of rooms, or height measured in both cm and inches■
  - ★ The columns of  $\mathbf{X}$  are nearly parallel so the data cannot tell us how to share the weight between them■
- Polynomial features  $x, x^2, \dots, x^9$  for  $x \in [0, 1]$  all look alike: the eigenvalues of  $\mathbf{X}^T \mathbf{X}$  span about 13 orders of magnitude■
- Features on very different scales (income in pounds and age in years)—easily fixed by standardising the features■

# Least Squares in Practice

- $\boldsymbol{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{y}$  is for understanding, not for computing
- Never compute  $(\mathbf{X}^\top \mathbf{X})^{-1}$ ; use a least squares solver such as `np.linalg.lstsq(X, y)`, which works with  $\mathbf{X}$  directly
- Forming  $\mathbf{X}^\top \mathbf{X}$  squares the ratio of largest to smallest eigenvalues:  $10^8$  becomes  $10^{16}$ , beyond the  $\sim 16$  significant figures of a double
- Standardise features (zero mean, unit variance) before fitting
- When  $m < p$ , `lstsq` returns the solution with the smallest  $\|\boldsymbol{w}\|$ —one version of the “simplest” solution (we will see why when we meet the SVD)

# Three Regimes

|            | $m > p$<br>over-constrained | $m = p$       | $m < p$<br>under-constrained |
|------------|-----------------------------|---------------|------------------------------|
| Exact fit? | usually not                 | yes           | yes, many                    |
| Unique?    | yes (usually)               | yes (usually) | no                           |
| Method     | least squares               | $w = X^{-1}y$ | pick “simplest”              |

- Real problems mix these: over-constrained in some directions and (nearly) under-constrained in others
- The nearly under-constrained directions make the problem ill-conditioned

# Summary

- Linear mappings are commonly used in machine learning algorithms such as regression■
- Systems can be over-constrained, under-constrained or ill-conditioned—often all at once■
- We will often meet the pseudo-inverse involving inverting  $\mathbf{X}^T \mathbf{X}$ ■
- Inverting an ill-conditioned  $\mathbf{X}^T \mathbf{X}$  amplifies noise in the data■